

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12 ASSESSMENT TASK 3

2004

Mathematics Extension 1

General Instructions

- Working time - 70 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks - 54

- Attempt Questions 1 – 6
- All questions are of equal value

Name : _____

Teacher : _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total

Question 1

Marks

i) Evaluate $\cos^{-1}(\cos \frac{2\pi}{3})$ 1

ii) Differentiate a) $\cos^2 3x$ 1

b) $\cos^{-1}(5x^2)$ 2

c) $\ln \left(\frac{x}{x+4} \right)$ 2

iii) a) Taking $x = 1$ as a first approximation to the root of 2

$$x - \tan^{-1}(2x) = 0$$

use Newton's method once to find a second approximation.
Write your answer correct to 2 significant figures.

b) Explain geometrically why using Newton's method for the equation in part a) would not have worked if we chose $x = 0.5$ as the first approximation. 1

Question 2 (Start a new page)

Marks

i) Solve $e^x = 8$ giving answer correct to 2 decimal places. 1

ii) Sketch $y = 4 \sin^{-1} 2x$ clearly showing the domain and range. 2

iii) Find a) $\int \frac{dx}{9+x}$ 1

b) $\int \frac{dx}{9+x^2}$ 1

iv) Solve $2 \ln x = \ln(3x+10)$ 3

v) Find the general solution of the equation $\tan x = \frac{1}{\sqrt{3}}$ 1

Question 3 (Start a new page)

Marks

- i) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$ 1
- ii) Find $\int \sin^2 5x \, dx$ 3
- iii) a) State the restricted domain of $y = -6x - x^2$ so that it will have an inverse function of the largest possible domain which will include the point $(-6, 0)$. 1
- b) Find the equation of this inverse function from part (a), writing y as the subject. 2
- c) Find any points of intersection of $y = -6x - x^2$ and the inverse found in part b). 2

Question 4 (Start a new page)

Marks

- i) a) Differentiate $x \cos 2x$ 1
- b) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} x \sin 2x \, dx$ 3
- ii) a) Find the stationary points of $y = x^2 \ln x$ and determine their nature. 3
- b) Sketch the curve $y = x^2 \ln x$ showing clearly its important features. 2

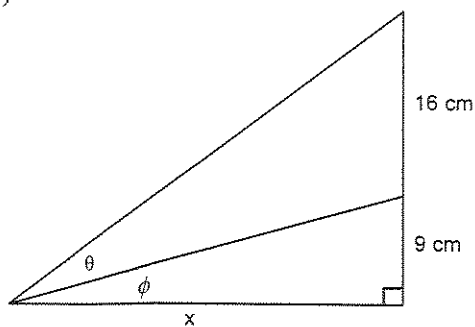
Question 5 (Start a new page)

Marks

i) Evaluate $\int_0^1 \frac{dx}{\sqrt{2-x^2}}$

3

ii)



a) Show that $\theta = \tan^{-1} \frac{25}{x} - \tan^{-1} \frac{9}{x}$

2

b) Use calculus to find the value of x that maximizes the value of θ .

4

Question 6 (Start a new page)

Marks

i) Find $\int \frac{dx}{\sqrt{4-25x^2}}$

2

ii) a) Show that the equation $x^3 - 5x + 3 = 0$ has a root between $x = 1$ and $x = 2$. Justify your answer.

2

b) Is the root of $x^3 - 5x + 3 = 0$ that lies between $x = 1$ and $x = 2$ closer to $x = 1$ or $x = 2$. Justify your answer.

1

iii) Solve $\tan^{-1} x + \tan^{-1} (1-x) = \tan^{-1} \frac{9}{7}$

4

QUESTION 1

$$\begin{aligned} \text{i)} \quad & \cos^{-1}\left(\cos \frac{2\pi}{3}\right) \\ &= \cos^{-1}\left(-\frac{1}{2}\right) \\ &= \frac{2\pi}{3} \end{aligned}$$

$$\begin{aligned} \text{ii) a.} \quad & \frac{d}{dx} (\cos^2 3x) \\ &= 2 \cos 3x \cdot -3 \sin 3x \\ &= -6 \cos 3x \sin 3x \end{aligned}$$

$$\begin{aligned} \text{b.} \quad & \frac{d}{dx} (\cos^{-1} 5x^2) \\ &= \frac{-10x}{\sqrt{1-25x^4}} \end{aligned}$$

$$\begin{aligned} \text{c.} \quad & \frac{d}{dx} \left(\ln \frac{x}{x+4} \right) \\ &= \frac{d}{dx} (\ln x - \ln(x+4)) \\ &= \frac{1}{x} - \frac{1}{x+4} \end{aligned}$$

$$\begin{aligned} \text{iii) a.} \quad & f(x) = x - \tan^{-1} 2x \\ & f'(x) = 1 - \frac{2}{1+4x^2} \end{aligned}$$

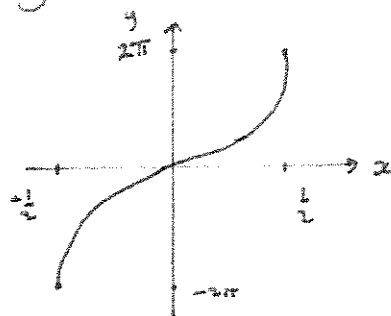
$$\begin{aligned} x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 1 - \frac{f(1)}{f'(1)} \\ &= 1 - \frac{1 - \tan^{-1} 2}{0.6} \\ &= 1.2 \end{aligned}$$

b. stationary pt at $x = 0.5$
 $\therefore$ tangent horizontal, never
 cuts x axis so no new approximation

QUESTION 2

$$\begin{aligned} \text{i)} \quad & e^x = 8 \\ & x = \ln 8 \\ &= 2.08 \end{aligned}$$

$$\text{ii)} \quad y = 4 \sin^{-1} 2x$$



$$\text{iii) a.} \quad \int \frac{dx}{9+x} = \ln(9+x) + C$$

$$\text{b.} \quad \int \frac{dx}{9+x^2} = \frac{1}{3} \tan^{-1} \frac{x}{3} + C$$

$$\begin{aligned} \text{iv)} \quad & 2 \ln x = \ln(3x+10) \\ & \ln x^2 = \ln(3x+10) \\ & x^2 = 3x+10 \\ & x^2 - 3x - 10 = 0 \\ & (x-5)(x+2) = 0 \\ & x = 5, -2 \\ & \text{but } x > 0 \\ & \therefore x = 5 \end{aligned}$$

$$\text{v)} \quad \tan x = \frac{1}{\sqrt{3}}$$

$$x = n\pi + \frac{\pi}{6}$$

where n is any integer

QUESTION 3

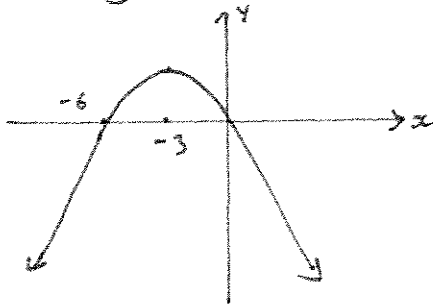
i) $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$

2. $\lim_{x \rightarrow 0} \frac{3 \cdot \frac{\sin 3x}{3x}}{3x}$
 $= 3$

ii) $\int \sin^2 5x \, dx$

$= \frac{1}{2} \int 1 - \cos 10x \, dx$
 $= \frac{1}{2} \left[x - \frac{1}{10} \sin 10x \right] + c$

iii) a. $y = -x(6+x)$



domain: $x \leq -3$

b. inverse

$x = -6y - y^2$
 $y^2 + 6y + x = 0$

$y = \frac{-6 \pm \sqrt{6^2 - 4 \cdot 1 \cdot x}}{2}$

$= \frac{-6 \pm \sqrt{36 - 4x}}{2}$

$= -3 \pm \sqrt{9 - x}$

take negative for required domain

$\therefore y = -3 - \sqrt{9 - x}$

c. intersect on $y = x$

$\therefore -6x - x^2 = x$

$x^2 + 7x = 0$

$x = 0, -7$

$x = 0$ not in domain

$\therefore \text{pt } (-7, -7)$

QUESTION 4

i) a. $\frac{d}{dx} (x \cos 2x) = \cos 2x - 2x \sin 2x$

b. $2x \sin 2x = \cos 2x - \frac{d}{dx} (x \cos 2x)$

$\therefore \int x \sin 2x \, dx = \frac{1}{2} \int \cos 2x - \frac{d}{dx} (x \cos 2x) \, dx$

$= \frac{1}{2} \left(\frac{1}{2} \sin 2x - x \cos 2x \right) + c$

$= \frac{1}{4} \sin 2x - \frac{1}{2} x \cos 2x + c$

ii) a. $y = x^2 \ln x \quad x > 0$

$\frac{dy}{dx} = 2x \ln x + x$

st. pts $y' = 0$

$2x \ln x + x = 0$

$x(2 \ln x + 1) = 0$

$x \neq 0, e^{-\frac{1}{2}}$

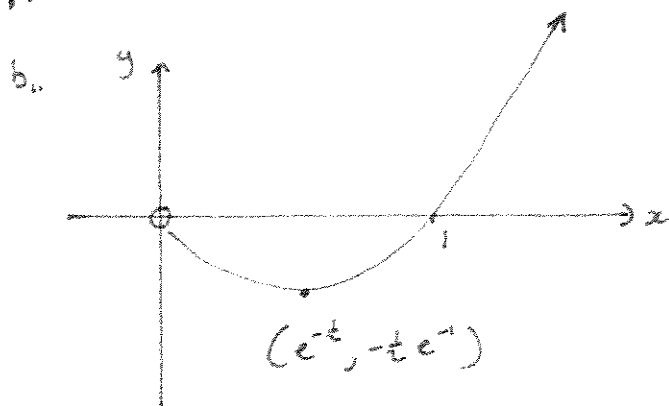
outside domain

when $x = e^{-\frac{1}{2}} \quad y = -\frac{1}{2} e^{-1}$

test $y'' = 2 \ln x + 3$

when $x = e^{-\frac{1}{2}}$

$y'' = +ve \Rightarrow \text{min at } (e^{-\frac{1}{2}}, -\frac{1}{2} e^{-1})$



QUESTION 5

i) $\int_0^1 \frac{dx}{\sqrt{2-x^2}}$

$$= \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1$$

$$= \sin^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} 0$$

$$= \frac{\pi}{4}$$

ii) a. $\tan \phi = \frac{9}{x}$

$$\tan(\phi + \theta) = \frac{25}{x}$$

$\therefore \phi = \tan^{-1} \frac{9}{x}$

$$\phi + \theta = \tan^{-1} \frac{25}{x}$$

$$\theta = \tan^{-1} \frac{25}{x} - \tan^{-1} \frac{9}{x}$$

b. $\theta = \tan^{-1} \frac{25}{x} - \tan^{-1} \frac{9}{x}$

$$\frac{d\theta}{dx} = \frac{-25}{x^2 + 625} + \frac{9}{x^2 + 81}$$

max/min $\frac{d\theta}{dx} = 0$

$$\frac{25}{x^2 + 625} = \frac{9}{x^2 + 81}$$

$$25x^2 + 2025 = 9x^2 + 5625$$

$$16x^2 = 3600$$

$$x^2 = 225$$

$$x = \pm 15$$

ignore -ve distance

$$\therefore x = 15$$

test

x	14	15	16
θ'	+	0	-

ve / -ve

$\therefore$ max when $x = 15$ cm

QUESTION 6

i) $\int \frac{dx}{\sqrt{4-25x^2}}$

$$= \frac{1}{5} \int \frac{dx}{\sqrt{\frac{4}{25} - x^2}}$$

$$= \frac{1}{5} \sin^{-1} \frac{5x}{2} + C$$

ii) a. $P(1) = -1$

$$P(2) = 1$$

opposite signs and curve

is continuous

$\therefore$ root between $x=1$ and $x=2$.

b. $P(1.5) = -1.125$

$\therefore$ root lies between $x=1.5$ and 2

$\therefore$ root closer to $x=2$.

$$\text{iii) } \tan^{-1} x + \tan^{-1}(1-x) = \tan^{-1} \frac{9}{7}$$

$$\text{let } \theta = \tan^{-1} x \Rightarrow \tan \theta = x$$

$$\alpha = \tan^{-1}(1-x) \Rightarrow \tan \alpha = 1-x$$

$$\therefore \tan^{-1} x + \tan^{-1}(1-x) = \tan^{-1} \frac{9}{7}$$

$$\therefore \tan(\tan^{-1} x + \tan^{-1}(1-x)) = \frac{9}{7}$$

$$\therefore \tan(\theta + \alpha) = \frac{9}{7}$$

$$\therefore \frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} = \frac{9}{7}$$

$$\therefore \frac{x + (1-x)}{1 - x(1-x)} = \frac{9}{7}$$

$$\frac{1}{1-x+x^2} = \frac{9}{7}$$

$$9 - 9x + 9x^2 = 7$$

$$9x^2 - 9x + 2 = 0$$

$$(3x-2)(3x-1) = 0$$

$$x = \frac{2}{3}, \frac{1}{3}$$